

PROPERTIES OF TWO-VARIABLE EXPANSIONS OF SCATTERING AMPLITUDE

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Two types of invariant expansions of scattering amplitude based on group $O(3, 1)$ that can be considered as generalizations of the partial wave analysis are studied. They are shown to be identical with the Fourier transform of the modified partial waves, the modification being given by integrating or differentiating the partial wave. We search for a simple physical use of expansion coefficients, the so-called Lorentz amplitudes, and find the connection between the poles of Lorentz amplitude and the high-energy behaviour of the partial wave; the relation between the Lorentz amplitude and the position of the Regge poles is precisely formulated.

1. INTRODUCTION

The scattering amplitude M_{fi} can be written in the form [1]

$$M_{fi} = \sum_n f_n(s, t, \dots) F_n$$

where F_n are constructed as invariants linearly dependent on wave amplitudes of all particles in question. The coefficients f_n (invariant amplitudes) depend on the Mandelstam variables $s, t, \dots$ and contain dynamical information about the scattering.

In the quantum electrodynamics we are in principle able to calculate f_n using the perturbation theory [1]. In the theory of strong interactions, we formulate dynamical postulates directly for the structure of these functions [2]. Two-variable expansions can be treated as a method of the latter approach. The functions are expanded with the use of an orthogonal and complete set of functions and we have to study the properties of expansion coefficients, Lorentz amplitudes (abbr. LA), instead of invariant amplitudes (review article [3] contains deeper analysis and list of basic references).

The set of generalized harmonic functions is determined by the choice of a group and its reduction to subgroups. The desired output of such a formulation is a simplification of some problems (fitting of data [4], formulation of hypotheses) and new understanding of old principles (analyticity, invariance, crossing etc.). The physical input of re-formulation is the interpretation of the group.

In the case of the scattering $a + b \rightarrow c + d$ with zero spin, it is possible to choose the reduction $O(3, 1) \supset O(3) \supset O(2)$, corresponding to S-expansion, or $O(3, 1) \supset O(2, 1) \supset O(2)$, expansion in H system [5]. The physical reasons for this choice are [3] as follows:

R1) $O(3, 1)$ is a group of motion of one of the particles on the mass shell, which is interpreted as a motion of the point in the physical region of s-channel in the Mandelstam plane.

R2) Two-variable expansions contain the partial wave expansions. The subgroups $O(3)$ and $O(2, 1)$ are also the little groups of the Poincaré group.

R3) S- and H-expansions in crossed channels are analytic continuations of each other [6].

R4) The formalism can be generalized for the case of non zero spins [7].

In our work we solve two problems in S- and H-system:

- 1) The formal simplification of the integral formulae.
- 2) The translation of asymptotics and poles of partial waves into the language of LA and vice versa.

The interest in point 1) is motivated by practical needs. Point 2) endeavours to connect the expansions with the usual theory of complex angular momentum.

2. GENERALIZED PW ANALYSIS

2.1. The formalism of S-expansion

According to Ref. [8] the following lemmas hold:

Lemma 1. Any function $F(u)$ of fourvector $u = (u_0, u_1, u_2, u_3)$

$$u = (\text{ch } a, \text{sh } a \sin \vartheta \cos \varphi, \text{sh } a \sin \vartheta \sin \varphi, \text{sh } a \cos \vartheta) \\ a \in (0, \infty), \quad \vartheta \in (0, \pi), \quad \varphi \in (0, 2\pi)$$

independent on variable φ , can be written as

$$(1) \quad F(u) = \sum_{l=1}^{\infty} (2l+1) a_l(a) P_l(\cos \vartheta)$$

with

$$(2) \quad a_l(a) = \frac{1}{\sqrt{(\text{sh } a)}} \int_0^{\infty} B_l(p) f_l(p) P_{-l-1/2+ip}^{-l-1/2}(\text{ch } a) dp$$

(notation of special functions see Ref. [9]), with inversion

$$(3) \quad a_l(a) = \frac{1}{2} \int_{-1}^1 d(\cos \vartheta) F(u) P_l(\cos \vartheta)$$

and

$$(4) \quad B_l(p) = f_l(-p) \int_0^{\infty} \sqrt{(\text{sh } a)} a_l(a) P_{-l-1/2+ip}^{-l-1/2}(\text{ch } a) \text{sh } a da$$

Here the abbreviation

$$f_l(p) = ip(ip-1) \dots (ip-l) = \Gamma(ip+1)/\Gamma(ip-l)$$

is introduced and the convergence of (1)–(4) is assumed (for the necessary and sufficient conditions see App. A).

Lemma 2. The mapping of the physical region – s-channel – in the Mandelstam plane into the hyperboloid $u^2 = 1$ has the form (masses equal 1/2; general form is given in [5])

$$\operatorname{ch} a = s^{1/2}, \quad \cos \vartheta = 1 + \frac{2t}{s-1}$$

If $f(s, t) = F(u)$ is an invariant amplitude, then (1) is the partial wave expansion for zero spin, the function (3) is partial wave (abbr. PW) and the relation (2) is the generalization of PW expansion. The Lorentz amplitude is defined in (4).

2.2. The convergence conditions and PW at threshold

Generalizing the properties of the Froissart-Gribov projection [10] the threshold behaviour of PW is postulated in the form

$$(5) \quad a_l(a) \sim a^{2l}(1 + O(a))$$

in the case $m_1 + m_2 = m_3 + m_4$ (e.g. equal masses) and in the form

$$(6) \quad a_l(a) \sim a^l(1 + O(a))$$

in other cases (i.e. non-coincidence of threshold and pseudothreshold). Both (5) and (6) are not in contradiction with the convergence condition (A1).

It is stated in Ref. [7] (for any spin), that PW has correct threshold behaviour. Such a strong statement is not valid for two reasons:

1) the condition (A1) admits non-physical behaviour (e.g. from $B_0(p) = p^{-1}$ we get $a_0(a) = (-\pi/2)^{1/2} (\operatorname{sh} a)^{-1}$), and

2) we can adopt different physical conditions, e.g. (5) or (6).

It could seem that we obtain (6) directly from (A5) (this is the method used in Ref. [7]), but the interchanging of limit with integration is justified only for LA, for which

$$\int_0^\infty B_l(p) f_l(p) p^{2N} dp < \infty$$

for any integer N . Then the function $a_l(a) \operatorname{sh} a^{-l}$ is analytic in the point $a = 0$ of a^2 -plane and difficulties arise in the case (5) for odd l .

The convergence condition (A2) leads to a restriction of the class of PW, which is too strong and excludes processes with cross-sections approaching non zero constants for high energy [3].

2.3. Lorentz amplitudes as a Fourier transform of a modified PW

Starting with recurrence relations for Legendre functions [9] we easily prove the formula

$$(7) \quad P_{-1/2+ip}^{-l-1/2}(\operatorname{ch} a) = \sqrt{\left(\frac{2}{\pi}\right)} \times \frac{\operatorname{sh} a^{l+1/2}}{|f_l(p)|^2} \left(-\frac{d}{d \operatorname{ch} a}\right)^{l+1} \cos pa$$

for integer l . Inserting it into (2) we get

$$a_l(a) = \operatorname{sh} a^l \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty dp \frac{B_l(p)}{f_l(-p)} \left(-\frac{d}{d \operatorname{ch} a}\right)^{l+1} \cos pa.$$

Interchanging the order of integration and differentiation (convergence of integrals does not become worse – use Appendix A) we obtain for PW the relation

$$(8) \quad a_l(a) = \operatorname{sh} a^l \left(-\frac{d}{d \operatorname{ch} a}\right)^l F_1(l, a)$$

where the new function is defined as

$$(9) \quad \operatorname{sh} a F_1(l, a) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty dp \sin pa \frac{p B_l(p)}{f_l(-p)}$$

and the inversion formula is valid

$$(10) \quad \frac{p B_l(p)}{f_l(-p)} = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty da \sin pa \operatorname{sh} a F_1(l, a).$$

Thus, the simple formulae (8), (9) and (10) are equivalent to (2) and (4) for physical amplitudes.

Another simple form of integral transformations (2) and (4) is obtained by inserting the function (7) into the integral (4). Supposing the physical threshold behaviour (5) or (6) we integrate per partes and have for the function

$$(11) \quad F_2(l, a) = \left(-\frac{d}{d \operatorname{ch} a}\right)^l a_l(a) \operatorname{sh} a^{l+1}$$

Fourier integral transformation

$$(12) \quad F_2(l, a) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty dp \sin pa f_l(p) B_l(p) p^{-1}$$

with inversion

$$(13) \quad f_l(p) B_l(p) p^{-1} = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty da \sin pa F_2(l, a).$$

For s-wave ($l = 0$) we have the relations

$$\operatorname{sh} a F_1(0, a) = F_2(0, a) = \operatorname{sh} a a_0(a)$$

Therefore, we shall call the functions $F_1(l, a)$ and $F_2(l, a)$ the modified PW.

2.4. Asymptotics of PW

Let us suppose that in the lower half of p -plane the function $B_l(p)$ is meromorphic and increasing in the infinity not faster than exponentially. This condition is not very restrictive one, if we believe in the reasonable properties of LA. It can be treated as a physical model.

Having for $a > \ln 2/2$ (see App. A) the expansion

$$(14) \quad \operatorname{sh} a a_l(a) = \frac{(-1)^{l+1}}{\sqrt{(2\pi)}} \sum_{k=0}^l \frac{(l+k)! \exp(-2ak)}{(l-k)! k! (1 - \exp(-2a))^k} \times \\ \times \int_{-\infty}^{\infty} \frac{B_l(p) e^{-ipa}}{(ip+1) \dots (ip+k)} dp$$

and applying the residuum theorem [11] for $a > \text{const.}$ we can substitute the sum over residua of LA $B_l(p)$ for integral in (2) or (14). For high energies, the asymptotics of PW (14) is equal to Fourier transform of LA

$$\operatorname{sh} a a_l(a) = \int_{-\infty}^{\infty} B_l(p) e^{-ipa} dp \left(\frac{(-1)^{l+1}}{\sqrt{(2\pi)}} + O(\exp(-2a)) \right)$$

and the dominant contribution comes from the pole of LA nearest to the real axis (cf. Regge poles and their role.) Shifting the integration path $p \rightarrow p + i\delta$ in (14) we would obtain the additional increase of PW generated by poles in the upper half of the p -plane, but it is connected with the transition to non-unitary representations of the group $O(3, 1)$ [3].

3. REGGE POLES AND LORENTZ AMPLITUDES

3.1. The formalism of H-expansion

According to Ref. [8] two lemmas hold:

Lemma 3. The function $F(u)$ of fourvector

$$u = (\operatorname{vh} \alpha \operatorname{ch} \beta, \operatorname{ch} \alpha \operatorname{sh} \beta \cos \varphi, \operatorname{ch} \alpha \operatorname{sh} \beta \sin \varphi, \operatorname{sh} \alpha) \\ \alpha \in (-\infty, \infty), \quad \beta \in (0, \infty), \quad \varphi \in (0, 2\pi),$$

independent on φ , can be expanded as

$$(15) \quad F(u) = -\frac{1}{2i} \int_{-1/2-i\infty}^{-1/2+i\infty} dl \frac{a(l, \alpha)}{\sin \pi l} P_l(\operatorname{ch} \beta)$$

where

$$(16) \quad a(l, \alpha) = -\cos \pi l \int_0^\infty F(u) P_l(\operatorname{ch} \beta) \operatorname{sh} \beta \, d\beta$$

Let us divide function (16) into two parts

$$a(l, \alpha) = a^{(+)}(l, \alpha) + a^{(-)}(l, \alpha), \quad a^{(\pm)}(l, \alpha) = \pm a^{(\pm)}(l, -\alpha)$$

Using notation (B2) we have

$$(17) \quad a^{(\pm)}(l, \alpha) = \frac{-\cos \pi l}{8\pi \operatorname{ch} \alpha \cdot \sqrt{2}} \int_{-\infty}^{\infty} A^{(\pm)}(p, l) \frac{\Gamma(ip-l) \Gamma(ip+l+1)}{\Gamma(ip+1)} \times \\ \times 2V^{(\pm)}(-p, l, \alpha) p^2 \, dp$$

where

$$(18) \quad A^{(\pm)}(p, l) = \frac{-\sqrt{(2)} \Gamma(-ip-l) \Gamma(-ip+l+1)}{\cos \pi l \Gamma(-ip+1)} \times \\ \times \int_{-\infty}^{\infty} 2V^{(\pm)}(p, l, \alpha) a^{(\pm)}(l, \alpha) \operatorname{ch} \alpha \, d\alpha$$

The convergence of integrals is assumed.

Lemma 4. The mapping of the s-channel physical region in the Mandelstam plane into the hyperboloid $u^2 = 1$ has the form

$$\operatorname{sh} \alpha = (-t)^{1/2}, \quad \operatorname{ch} \beta = -1 - \frac{2s}{t-1}, \quad \alpha \in (0, \infty), \quad \beta \in (0, \infty)$$

if $m = 1/2$ (see [5]). The relation (15) is the background integral of (1), after the crossing and Sommerfeld-Watson transformation [10, 12], if invariant amplitude $F(u) < s^{-1/2}$, $s \rightarrow \infty$. Then (17) and (18) define PW and LA, respectively.

Let us adopt the restriction $a^{(-)}(l, \alpha) \equiv 0$. This relation is valid for equal masses and real α ; in this case is $\alpha \rightarrow -\alpha$ a Lorentz rotation in space. This restriction will simplify our calculations, and the method can be used for $a^{(-)}(l, \alpha) \neq 0$ as well.

3.2. Regge poles

The importance of Regge poles need not be stressed. The expression (17) of PW enables us to transform the information about the position of Regge poles into LA, if the formula (17) is continued in the complex α -plane. This continuation is made in

Ref. [12] and it is shown there that a certain type of asymptotics of LA produces moving singularity of PW — i.e. the Regge pole. We assert

1) The argumentation of the article [12] about correspondence between Regge poles and Lorentz asymptotics is not valid (App. C).

2) The correct result of the example with Coulomb potential is obtained by accidental transition to another exact integral transformation (Appendix C).

3) A modified PW, function $g(l, \alpha)$, can be introduced so that

a) $g(l, \alpha)$ is the Fourier transform of the product of LA and a factor. This factor is equal to 1 in the limit $p \rightarrow \infty$.

b) $g(l, \alpha)$ approximates PW in the vicinity of the Regge poles (see following subsection).

3.3. Lorentz amplitude as a Fourier transform of a modified PW

Let us use the PW of the type (B8). This can be done without loss of generality for a real α . We shall not tackle here the subtle questions of analytic continuation in α .

The Lorentz amplitude (B9) can be written in the form (cf. [9], formula no. 8.380.10) analogous to (C2)

$$(19) \quad A^{(+)}(p, l) = \frac{L_1(p, l) \sqrt{2}}{(ip/2) \cos \pi l} \int_{-\infty}^{\infty} d\alpha \cos p\alpha \operatorname{ch} \alpha g(l, \alpha)$$

where $g(l, \alpha)$, the modified PW, is defined as

$$(20) \quad \operatorname{ch} \alpha g(l, \alpha) = \int_G dr \frac{1}{\operatorname{ch}^r a} D_l(r) {}_2F_1(-l, l+1, r, \frac{1}{2})$$

and is equal to

$$(21) \quad \operatorname{ch} \alpha g(l, \alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dp \cos p\alpha A^{(+)}(p, l) \frac{(ip/2) \cos \pi l}{L_1(p, l) \sqrt{2}}$$

(cf. (C1)). The resemblance of (20) and (B8) enables us to avoid the use of $D_l(r)$ in the following way. We write the hypergeometric function in (20) as a sum ([9], no. 9.100), beta function containing the dependence on parameter r being given by the definition no. 8.830 in [9]. The final result — after some trivial manipulations with [9] — is

$$(22) \quad \operatorname{ch}^2 \alpha g(l, \alpha) = \operatorname{ch}^2 \alpha a(l, \alpha) + \int_{t=1}^{t=0} \operatorname{ch}^2 \alpha' a(l, \alpha') d(P_l(t))$$

where

$$\operatorname{ch} \alpha' = (\operatorname{ch} \alpha)/t.$$

The weight function $P_l(t)$ is a bounded and decreasing positive Legendre function with derivative of analogous properties. Thus, the integral in (22) converges, unless

PW has singularities on the integration path. In the vicinity of the Regge pole, the difference between $g(l, \alpha)$ and PW is less singular than the pole term. This means that the modified PW contains information about the position of the Regge pole but not about the residuum. The singularity is a branching point.

4. CONCLUSIONS

1) Two-variable expansions for zero spin can be used for any invariant amplitude (any spin); they can help in the investigation of analytic properties of amplitudes and model Lorentz amplitudes can be compared with the experiment. The interpretation (R2) (see Introduction) is lost, however. It is re-established in another formalism, which is much more complicated [7].

2) The convergence conditions necessary for applications of S-expansions are formulated in Appendix A. The threshold behaviour of PW is discussed in subsection 2.2. It is shown that the physical threshold conditions are not implicit in the formalism, in contradiction with Ref. [7].

3) Lorentz amplitude is expanded as Fourier transform of modified PW (functions (10), (13) and (21)). There arises the possibility to work directly with those functions. For s-wave, the modified PW is equal to PW.

4) Subsection 2.4 is an example of a direct use of Lorentz amplitudes — poles in LA dominate the high energy behaviour of PW. This conclusion is valid also for the H-expansion, i.e. for non-physical energies (see App. B).

5) Integral transformation in H-expansion is complicated. The important simplification is discussed in App. C, that polemizes with the work of Ref. [12] in the matter of Regge poles. It is shown that only a few (in energy plane non preferred) poles influence the Lorentz asymptotics and that only these poles are reproduced by Lorentz asymptotics.

6) Exact results of App. B enable us to reproduce the position of all the Regge poles using modified PW, i.e., the Fourier transform of Lorentz amplitude multiplied by the known function. Thus, not only the asymptotic region in the p-plane is relevant to the knowledge of the Regge poles.

APPENDIX A

Let us write the integral (2) in the form

$$a_l(a) = \int_0^\infty \dots = \int_0^\varepsilon + \int_\varepsilon^N + \int_N^\infty = J_1(\varepsilon) + J_2 + J_3(N)$$

and the integral (4) similarly as

$$B_l(p) = \int_0^\infty \dots = \int_0^\varepsilon + \int_\varepsilon^N + \int_N^\infty = J_4(\varepsilon) + J_5 + J_6(N)$$

Let us suppose that $J_2 < \infty$, $J_5 < \infty$ for any $N > \varepsilon > 0$. (suffice to assume that both PW and LA are continuous).

Let us write

$$(A1) \quad a_l(a) \sim a^{-l-3+\eta}(1 + O(a)) \quad \text{for } a \rightarrow 0,$$

$$(A2) \quad a_l(a) \text{ ch } a \rightarrow 0 \quad \text{for } a \rightarrow \infty,$$

$$(A3) \quad B_l(p) \rightarrow 0 \quad \text{for } p \rightarrow \infty,$$

$$(A4) \quad B_l(p) \sim p^{-2+\eta}(1 + O(p)) \quad \text{for } p \rightarrow 0.$$

If (A1) is valid, then $J_4(\varepsilon) < \infty$ if and only if $\eta > 0$.

If (A2) is valid, then $J_6(N) < \infty$.

If (A3) is valid, then $J_3(N) < \infty$.

If (A4) holds, then $J_1(\varepsilon) < \infty$ if and only if $\eta > 0$.

The proof of these statements is trivial and is based on relations no. 8.702 and 8.723 in [9] (l integer):

$$(A5) \quad P_{-1/2+i p}^{-l-1/2}(\text{ch } a) = \frac{\text{th}(a/2)^{l+1/2}}{\Gamma(l + \frac{3}{2})} {}_2F_1\left(\frac{1}{2} - ip, \frac{1}{2} + ip, l + \frac{3}{2}, -\text{sh}(a/2)^2\right)$$

$$(A6) \quad P_{1/2+i p}^{-l-1/2}(\text{ch } a) = \frac{(-1)^{l+1} e^{-i p a}}{f_l(p) (2\pi \text{sh } a)^{1/2}} {}_2F_1\left(-l, l + 1, 1 + ip, \frac{1}{1 - e^{2a}}\right) + \\ + \text{h.c.}, \quad a < (\ln 2)/2.$$

APPENDIX B

Let us introduce the notation

$$(B1) \quad 2V^{(\pm)}(p, l, \alpha) = P_l^i p(-\text{th } \alpha) \pm P_l^i p(\text{th } \alpha).$$

According to equation 8.775.1 in [9]

$$(B2) \quad 2V^{(\pm)}(p, l, \alpha) = \frac{(1 - \xi^2)^{-ip/2} 2^{ip+1} \sqrt{(\pi)} {}_2F_1\left(\frac{b^-}{2}, \frac{b^+}{2}, \frac{1}{2}, \xi^2\right)}{\Gamma\left(\frac{1+b^-}{2}\right) \Gamma\left(\frac{1+b^+}{2}\right)}$$

where

$$(B3) \quad l = -\frac{1}{2} + iq, \quad \text{th } \alpha = \xi, \quad b^\pm = \frac{1}{2} - ip \pm iq.$$

Let us rewrite (18) in the form

$$(B4) \quad A^+(p, l) = \frac{\sqrt{(2\pi)}}{(ip/2) \cos \pi l} L_1(p, l) I(p, l)$$

where

$$(B5) \quad I(p, l) = 2 \int_0^1 (1 - \xi^2)^{-3/2} d\xi a^{(+)}(l, \alpha) \times \\ \times \left\{ (1 - \xi^2)^{-ip/2} \Gamma(ip) \frac{{}_2F_1\left(\frac{b^-}{2}, \frac{b^+}{2}, 1 - ip, 1 - \xi^2\right)}{\Gamma\left[\frac{(b^-)^*}{2}\right] \Gamma\left[\frac{(b^+)^*}{2}\right]} + h.c. \right\}$$

is symmetric with respect to reflexion $p \rightarrow -p$ and

$$(B6) \quad L_1(p, l) = ({}_2F_1(-l, l+1, -ip, \frac{1}{2}) = \\ = \frac{\Gamma[(-ip-l)/2] \Gamma[(-ip+l+1)/2]}{\Gamma[(-ip)/2] \Gamma[(-ip+1)/2]} = \prod_{k=0}^{\infty} \frac{(2k + \frac{1}{2} - ip)^2 - \frac{1}{4}}{(2k + \frac{1}{2} - ip)^2 + q^2}$$

denotes the function that changes to its hermitian conjugate while $p \rightarrow -p$.

The convergence condition for integral (B5) is easily derived in the form

$$(B7) \quad a^{(+)}(l, \alpha) \sim \text{ch } \alpha^{-1-\eta}, \quad \alpha \rightarrow \infty, \quad \text{Re } \eta > 0$$

Let us consider the class of functions of the type

$$(B8) \quad a^{(+)}(l, \alpha) = \frac{1}{\text{ch } \alpha} \int_G \frac{1}{\text{ch } \alpha^r} D_l(r) dr$$

where $D_l(r)$ is (generalized) function defined on the half-plane $\text{Re } r > 0$, in order to satisfy (B7). Inserting (B8) into (18), interchanging the order of integrals and integrating term by term we obtain

$$(B9) \quad A^+(p, l) = \frac{L_1(p, l) \sqrt{(2\pi)}}{(ip/2) \cos \pi l} \int_G I(p, l, r) D_l(r) dr$$

where

$$(B10) \quad I(p, l, r) = \frac{\Gamma[(r-ip)/2] \Gamma[(r+ip)/2]}{\Gamma[(r+\frac{1}{2}-iq)/2] \Gamma[(r+\frac{1}{2}+iq)/2]}$$

We shall not specify the form of $D_l(r)$, even if it were interesting for application to treat (B8) as a sum, Laplace transform etc.

There is an interesting phenomenon. Approximation of $D_l(r)$ by delta function, which reproduces the asymptotic behaviour of (B8) for $\alpha \rightarrow \infty$, leads to the pole in LA (from gamma function in (B10)) in analogy with results of subsection 2.4.

APPENDIX C

Let us repeat the argumentation of Ref. [12]. We divide the integral (17) into a sum of three integrals J_1, J_2, J_3 in intervals $p \in (-\infty, -N), (-N, N), (N, \infty)$, respectively. In the vicinity of the Regge pole $J_1 + J_3 \gg J_2$. We use the asymptotic expansion of integrated functions in J_1, J_3 and inverse formula (18) and obtain

$$(C1) \quad \text{ch } \alpha a^{(+)}(l, \alpha) = \text{ch } \alpha J_2 + \frac{i \cos \pi l}{8\pi \sqrt{2}} \left(\int_{-\infty}^{-N} + \int_N^{\infty} \right) p A^+(p, l) \cos pa \, dp + O\left(\frac{1}{N}\right)$$

for PW and

$$(C2) \quad A^+(p, l) = \frac{\sqrt{2}}{(ip/2) \cos \pi l} \int_{-\infty}^{\infty} d\alpha \text{ch } \alpha a^{(+)}(l, \alpha) \cos p\alpha \left(1 + O\left(\frac{1}{p}\right) \right)$$

for asymptotics of LA. Using the residuum theorem we get

$$(C3) \quad A^+(p, l) = \sum_n f(\alpha_n) \exp(ip \text{Re } \alpha_n) \exp(-p |\text{Im } \alpha_n|) \left(1 + O\left(\frac{1}{p}\right) \right)$$

for well-behaved PW.

Our comments:

1) The expression (C3) needs further rearrangement because $p^{-1} \gg \exp(-p)$ for $p \rightarrow \infty$. The sum is reduced to that of the terms with $|\text{Im } \alpha_n| = \text{minimum}$, because the other terms are less than the neglected part of asymptotic expansion.

2) If we neglect the higher terms in (C2), (C1) and take $N \rightarrow 0$ in (C1) – it is not trivial since $1/N \rightarrow \infty, 1/p \rightarrow \infty$ and approximations cease to be valid – we obtain (accidentally, but exactly) the Fourier transformation formulae. That is why the results of Ref. [12] are correct in the case of Coulomb potential.

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